The docking score implemented in this repository

Differentiable ZDOCK — `src/zdock`

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Abstract

This note states, in closed form, the scoring function evaluated by `zdock.score.docking_score_elec` and maximised by `zdock.search.docking_search`, and maps every symbol onto the primary literature: the pairwise shape-complementarity term of Chen & Weng (2003) [1], the interface atomic-contact-energy pair potential of Mintseris *et al.* (2007) [3], and the combination with electrostatics of Chen *et al.* (2003) [2], with the earlier grid-based formulation of Chen & Weng (2002) [4] given only where it explains a symbol that survives in the code. It also records where the implementation deviates from the papers, and why.

1 Total score

Let R denote the receptor (held fixed at the origin) and L the ligand, placed by a rigid transform $(q, \mathbf{t})$: a rotation q drawn from a finite quaternion set, followed by a translation $\mathbf{t}$ on the FFT lattice. All three terms are evaluated by scattering atoms onto a Cartesian grid of spacing h and contracting grid functions, so that the translational scan is a single FFT correlation per rotation — the construction of Katchalski-Katzir *et al.* [6], used by every ZDOCK version [4, 1, 2]. The total is

$$S(q,\mathbf{t}) \;=\; \alpha\, S_{\mathrm{PSC}}(q,\mathbf{t}) \;+\; S_{\mathrm{IFACE}}(q,\mathbf{t}) \;+\; \beta\, S_{\mathrm{ELEC}}(q,\mathbf{t}), \tag{1}$$

and **higher is better**: `docking_search` returns the top-N poses by S. Section 5 explains why keeping all three terms in a “higher is better” convention requires an explicit sign flip on S_{IFACE}.

Grid. $h = 1.2$ Å throughout, the value used by every ZDOCK paper (“A grid spacing of 1.2 Å is used throughout this study” [1]). An atom is a *surface* atom if its solvent-accessible area, computed with a 1.4 Å water probe, exceeds 1 Å^2; otherwise it is a *core* atom ([1], Methods; same definition in [4]).

2 Shape complementarity: PSC

2.1 Grid functions

Following Chen & Weng [1], Eq. (3), each molecule is described by one complex grid function. Write $\mathcal{S}$ and $\mathcal{C}$ for the sets of surface and core atoms, r_j for the van der Waals radius of atom j, and $\mathbf{x}_c$ for the centre of grid cell c. Define the occupancy sets

$$\begin{aligned} \mathrm{surf}(c) \;&\Longleftrightarrow\; \exists j \in \mathcal{S} : \|\mathbf{x}_c - \mathbf{x}_j\| < r_j, & (2)\\ \mathrm{core}(c) \;&\Longleftrightarrow\; \exists j \in \mathcal{C} : \|\mathbf{x}_c - \mathbf{x}_j\| < r_j, & (3) \end{aligned}$$

with surf taking precedence (“the solvent excluding surface layer … is defined by the grid points corresponding to surface atoms; *all other* grid points corresponding to any core atoms are in the protein core" [1, 2]), and let $\mathrm{open}(c) \iff \neg\,\mathrm{surf}(c) \wedge \neg\,\mathrm{core}(c)$. Note that PSC, unlike the earlier GSC, defines *no* solvent-accessible surface layer: "any grid point that does not correspond to an atom is in the open space" [1].

The imaginary channel is the same for both partners and encodes the clash penalty,

$$\mathrm{Im}[R_{\mathrm{PSC}}(c)] = \mathrm{Im}[L_{\mathrm{PSC}}(c)] = \begin{cases} \rho & \mathrm{surf}(c) \\ \rho^2 & \mathrm{core}(c) \\ 0 & \mathrm{open}(c), \end{cases} \tag{4}$$

while the real channel differs between the two and encodes the favourable atom-pair count,

$$\mathrm{Re}[R_{\mathrm{PSC}}(c)] = \begin{cases} \big|\{\, j \in R : \|\mathbf{x}_c - \mathbf{x}_j\| < D + r_j \,\}\big| & \mathrm{open}(c) \\ 0 & \text{otherwise}, \end{cases} \tag{5}$$

$$\mathrm{Re}[L_{\mathrm{PSC}}(c)] = \big|\{\, k \in L : c = \mathrm{nearest}(\mathbf{x}_k) \,\}\big|. \tag{6}$$

2.2 Score

Eq. (4) of [1] takes the *real part* of the cross-correlation only:

$$S_{\mathrm{PSC}}(\mathbf{t}) = \mathrm{Re}\Big[\textstyle\sum_c R_{\mathrm{PSC}}(c)\, L_{\mathrm{PSC}}(c+\mathbf{t})\Big] = \underbrace{\sum_c \mathrm{Re}[R]\,\mathrm{Re}[L]}_{\text{favourable}} - \underbrace{\sum_c \mathrm{Im}[R]\,\mathrm{Im}[L]}_{\text{clash penalty}}. \tag{7}$$

The first sum is exactly the number of receptor–ligand atom pairs within the cutoff, because $\mathrm{Re}[L]$ places one count at each ligand atom and $\mathrm{Re}[R]$ reports how many receptor atoms lie within $D + r_j$ of that point. The second sum contributes

overlap	value	$\rho = 3.5$
surface–surface	$-\rho^2$	-12.25
surface–core	$-\rho^3$	-42.88
core–core	$-\rho^4$	-150.06

so deeper interpenetration is punished super-linearly, "with a higher score indicating better shape complementarity" [1].

Caveat on ρ. These are *not* the penalties printed in the PSC paper, which states them explicitly: "a core–core, surface–core, or surface–surface grid point overlap results in a penalty of $-9\times9 = -81$, $-3\times9 = -27$, and $-3\times3 = -9$ respectively" [1], i.e. $\rho = 3$ against a favourable weight of $+1$ per pair. The $\rho = 3.5$ used here comes from Eq. (2) of [2], but there the favourable term is simultaneously raised to 1.334 per pair ("we decrease the PSC score for each atom pair from -1 to -1.334 … the PSC penalty is increased slightly to balance the increased favorable contribution by electrostatics"). The penalty:reward ratio is 9.0 in the first formulation and 9.18 in the second; the present code mixes $\rho = 3.5$ with a reward weight of 1, giving 12.25 — about 36% harsher than either published variant. Either set $\rho = 3$, or keep 3.5 and set the pair weight to 1.334.

Why this term dominates. On clash-free surface-placed decoys the bare pair count $\sum_{ij} n_{ij}$ separates near-native from non-native poses with a Mann–Whitney AUC of 0.998, i.e. it is by far the strongest single signal in the score. Omitting it — as this port did until the present revision — makes every contact a net penalty and drives the search towards poses that do not touch at all.

3 Desolvation: the IFACE pair potential

Mintseris *et al.* [3] replace ZDOCK's averaged ACE desolvation by a pairwise statistical potential over $k = 12$ optimised atom types. With n_{ij} the number of contacts between receptor atoms of type i and ligand atoms of type j, their Eq. (11) and Eq. (14) read

$$e_{ij} = \ln \frac{n_{ij}}{c_{\text{surface-fraction};ij}}, \qquad E_{\text{IFACE}} = \sum_{i=1}^{k} \sum_{j=1}^{k} e_{ij}\, n_{ij}. \tag{8}$$

On the grid (their Eq. (13)) the receptor side pre-multiplies the potential,

$$W_i(c) = \sum_j e_{ij}\, H_j(c), \qquad H_j(c) = \big|\{\, m \in R \text{ of type } j : \|\mathbf{x}_c - \mathbf{x}_m\| < 6\,\text{Å} \,\}\big|, \tag{9}$$

the 6 Å contact sphere being the one that defines n_{ij} in [3], Eqs. (8)–(9), and the ligand side is the per-type occupancy indicator of their Eq. (12), $L_i(c) \in \{0, 1\}$, so that

$$S_{\text{IFACE}}(\mathbf{t}) = \sigma \sum_i \sum_c W_i(c)\, L_i(c + \mathbf{t}), \qquad \sigma = -1. \tag{10}$$

4 Electrostatics

Chen & Weng [4] write the electrostatic energy as a correlation between the receptor's electric potential and the ligand's atomic charges, using CHARMM19 partial charges, with grid points in the receptor core assigned zero potential ”to avoid the contributions from non-physical receptor-core/ligand contacts”; Chen *et al.* [2], Eq. (2), carry the same term into ZDOCK through the imaginary channel. Here it is kept as a separate real correlation:

$$V(c) = \sum_{m \in R} \frac{q_m}{\|\mathbf{x}_c - \mathbf{x}_m\|} \quad (\|\cdot\| < 8\,\text{Å},\ \text{open}(c)), \qquad S_{\text{ELEC}}(\mathbf{t}) = \sum_c V(c)\, Q_L(c + \mathbf{t}). \tag{11}$$

5 Sign conventions

The two ingredient papers disagree about which direction is favourable, and Chen *et al.* [2] say so explicitly:

> “positive PSC scores indicate good shape complementarity … whereas ACE scores can be positive (unfavorable) or negative (favorable) … To make these two scores compatible, we flip the signs of the PSC scores.”

They resolve it by flipping PSC and *minimising*. Because the FFT search here selects poses by `topk`, this implementation flips the other term instead: $\sigma = -1$ in Eq. (10) turns the ACE-convention pair potential into a “higher is better” quantity, and every term in Eq. (1) then agrees with S_{PSC}. Omitting this flip makes S_{IFACE} anti-correlated with pose quality (measured AUC 0.127; the negated term gives 0.873), and since $|S_{\text{IFACE}}| \approx |S_{\text{PSC}}|$ it then dominates the total.

6 Parameters

symbol	meaning	initial value	source
h	grid spacing	1.2 Å	[1], Methods
D	PSC pair cutoff offset	3.6 Å	[1], "Optimizing PSC"
ρ	PSC clash scale	3.5	[2], Eq. (2)
α	weight on S_{PSC}	1.0	see below
β	weight on S_{ELEC}	3.0	[2], p. 82
e_{ij}	12×12 pair potential	IFACE table	[3], Eq. (11)
q_m	partial charges (11 types)	CHARMM19	[4], "Electrostatics"
σ	pair-term sign	-1	[2], p. 81

ρ, α, β, e_{ij} and q_m are exposed as differentiable tensors and can be optimised; the values above are the initialisation. D **cannot**: it enters only through the hard distance test $\|\mathbf{x}_c - \mathbf{x}_j\| < r_j + D$, whose output is a boolean selection, so $\partial S/\partial D \equiv 0$ (verified: the score changes from -11138.30 at $D = 3.5$ to -11111.30 at $D = 3.6$ but autograd returns `None` and every finite difference below $h = 10^{-2}$ is exactly zero). D must be tuned by grid search, or the cutoff replaced by a smooth switching function. Note also that q_m is listed as CHARMM19 above but the table currently in the code is not CHARMM19 (the backbone N even has the opposite sign); see the audit in `EXPERIMENT_REPORT.md` §5.7.7. Note that α has no counterpart in Eq. (2) of [2], where the scale of the PSC term is fixed by ρ and the term enters with unit weight; the $\alpha = 0.01$ carried in earlier revisions of this code belongs to the *grid*-based formulation of [4], Eq. (6), whose second term is an averaged ACE energy on a kcal/mol scale rather than a pair count.

7 Where each equation comes from

here	quantity	source	source equation
(1)	$S = \alpha S_{\text{PSC}} + S_{\text{IFACE}} + \beta S_{\text{ELEC}}$	[2]	Eq. (2)
(4)	clash channel ρ, ρ^2	[1]	Eq. (3), Im
(5)	receptor pair-count channel	[1]	Eq. (3), Re[R]
(6)	ligand nearest-grid channel	[1]	Eq. (3), Re[L]
(7)	$S_{\text{PSC}} = \text{Re}[R \cdot L]$	[1]	Eq. (4)
(8)	e_{ij}, E_{IFACE}	[3]	Eqs. (11), (14)
(10)	grid form of the pair sum	[3]	Eqs. (12), (13)
$\sigma = -1$	sign reconciliation	[2]	p. 81
$D = 3.6$ Å	pair cutoff offset	[1]	"Optimizing PSC"
$\rho = 3.5$	clash scale	[2]	Eq. (2)
$\beta = 3$	electrostatics weight	[2]	p. 82
$h = 1.2$ Å	grid spacing	[1]	Methods

The four ZDOCK papers are in `ref/` of this repository (file names in the bibliography); [5, 6, 8] are cited for provenance and are not needed to run the code.

8 Deviations from the papers

1. **Rotational sampling.** ZDOCK uses evenly distributed Euler angle sets — 1800 orientations at $\Delta = 20°$, 54 000 at 6°, 180 000 at 4° ([1], "Rotational Sampling"). This implementation draws uniform random quaternions, so for the same count the worst-case angular error is larger. Finer is not automatically better: [1] report that "finer angular

intervals tend to rank the best hits lower than coarser angular intervals", $\Delta = 4°$ giving the *lowest* success rate for $N_P > 100$.

2. **Ligand occupancy in** S_{IFACE}**.** Eq. (12) of [3] is a binary indicator (their Eq. (14) uses "$\cong$", acknowledging that the grid form only approximates $\sum_{ij} e_{ij} n_{ij}$), which under-counts when two ligand atoms of the same type share a cell. That is negligible at $h = 1.2$ Å (cell volume 1.7 Å^3) but not at coarser spacings; the quantity is 37% smaller at $h = 3.0$ Å.

3. **Cell-volume normalisation.** The clash channel counts grid cells, so it scales as h^{-3}. `sc_cell_volume_factor` rescales it to the 1.2 Å reference, making the term spacing-invariant; it is the identity at the default spacing.

4. **DockQ.** Pose quality is scored against [8] by an atom-level approximation without residue or backbone metadata, and without symmetry handling, so the 0.23 acceptance threshold is not identical to canonical CAPRI. Ligand RMSD is reported alongside as a cross-check.

9 Rotational sampling

ZDOCK does not sample rotations at random: "Evenly distributed Euler angles are used for the rotational search. The Euler angle sets … have been obtained from Dr. Julie C. Mitchell. An angle set is equivalent to a uniformly distributed set of points on a projective sphere, which ensures that minimal orientations are required to cover the entire rotational space. The angular distance between any orientation and its nearest orientation … is Δ or smaller" [1]. The published sets are not distributable, so this implementation generates an equivalent grid with the Hopf-fibration construction of Yershova *et al.* [7] — the same Mitchell as the ZDOCK acknowledgement. Writing $S^3 = S^2 \times S^1$,

$$q(\theta, \phi, \psi) = \left(\cos\tfrac{\theta}{2}\cos\tfrac{\psi}{2},\ \cos\tfrac{\theta}{2}\sin\tfrac{\psi}{2},\ \sin\tfrac{\theta}{2}\cos(\phi + \tfrac{\psi}{2}),\ \sin\tfrac{\theta}{2}\sin(\phi + \tfrac{\psi}{2})\right), \tag{12}$$

with (θ, ϕ) the centres of the $12\,n_{\text{side}}^2$ equal-area HEALPix pixels and ψ an even grid of $6\,n_{\text{side}}$ points on the circle.

What matters is the *covering radius*: the largest angle any orientation can be from the grid bounds how far a rigid ligand must be rotated away from its true pose, and therefore how much clash penalty the discretisation forces onto the correct answer. The **max** column below is a sample maximum over 4000 random probes and is therefore a downward-biased *lower bound* on the covering radius (a 2M-probe reference with local ascent gives $\geq 18.5°$, $\geq 13.9°$ and $\geq 6.2°$ for the three Hopf rows). The mean and p95 columns are converged. Note the Δ quoted by [1] is a *guaranteed* bound, so it is not measured the same way; packing efficiency is the like-for-like comparison, and there the two agree (0.25–0.31 for the paper's sets, 0.29 for the Hopf grids).

rotation set	N	mean	p95	max
Hopf $n_{\text{side}} = 3$	1 944	9.4°	13.2°	**16.8**°
uniform random	1 900	11.1°	17.9°	25.9°
Hopf $n_{\text{side}} = 4$	4 608	7.0°	9.9°	**12.2**°
uniform random	4 608	8.2°	13.4°	20.6°
Hopf $n_{\text{side}} = 9$	52 488	3.1°	4.4°	**5.5**°
uniform random	52 488	3.6°	5.8°	8.8°

At essentially equal point count (1 944 vs 1 900) the Hopf grid cuts the worst case by 35%; that relative figure is robust, since both sides are estimated the same way (true values 18.53° vs 28.33°). The tail improves far more than the mean. $n_{\text{side}} = 3$ corresponds to $\Delta \gtrsim 18.5°$, comparable to the paper's "1800 orientations at $\Delta = 20°$" in packing efficiency.

Translations kept per rotation. "Only the best translational orientation is kept for every rotational orientation. We used to keep the top 10 … keeping only the best one helped to remove false positives without affecting the ranking of the best hit" [1]. The default here follows the paper (1). This is non-maximum suppression rather than a pure top-K over poses, so success@K under the two conventions is *not* the same quantity and the two must not be compared directly.

10 Measured effect

Interface-deleaked PINDER complexes, default parameters (no training), $h = 1.2$ Å, top-2000 search output.

Correction (2026-07-25). An earlier version of this caption claimed *nothing injected into the candidate set and no native-informed rotations* for the whole table. That is true only of the last two rows. Rows 1–3 were produced by `validate_grid_spacing.py` at its then-default `--n-cone 400`, i.e. with a 25° cone around the native orientation q^* mixed into the rotation set, which leaks the answer and inflates both recall and the ceiling; rows 1–2 are also over *20* complexes, not 12. They are upper bounds on the corresponding cone-free numbers and are marked accordingly. The script default is now 0.

implementation	recall	best DockQ	s@1	s@10	s@100
clash only, $h = 3.0$ Å *(cone, n=20)*	0.0%	0.107	0%	0%	0%
clash only, $h = 1.2$ Å *(cone, n=20)*	0.0%	0.066	0%	0%	0%
+ pair-term sign flip *(cone, n=12)*	16.7%	0.116	0%	0%	0%
full PSC, uniform rot. *(cone-free, n=12)*	58%	0.398	8%	17%	42%
full PSC, Hopf grid, 1 trans/rot	**75%**	**0.53**	**8%**	**17%**	**25%**

Recall — whether the search surfaces a near-native pose at all — goes from 0% to 75%, but success@1 stays at 8% (1 of 12) in every configuration. The answer is now returned; it is not ranked first.

11 Why the correct pose loses

Decomposing the score of three poses per complex — the exact native (q^*, t^*), the best *reachable* pose (nearest sampled rotation at the lattice-snapped t^*), and the pose the search actually ranked first — separates the two failure modes. Grouping by whether the search recovered a near-native pose anywhere in its output:

	n	αS_{fav}	$-\alpha S_{\text{clash}}$	S_{IFACE}	total
search top-1 minus best reachable (positive = prefers the wrong pose)					
recall > 0	9	+102	**+742**	+139	+982
recall $= 0$	3	**+706**	−180	**+366**	+891
best reachable minus exact native (what discretisation costs)					
recall > 0	9	+109	**−966**	−8	−865
recall $= 0$	3	+17	−72	−17	−71

For the nine complexes where the answer is found, essentially the entire discretisation cost is the clash penalty. Snapping the native orientation to the nearest grid point (9° on average) makes a *rigid* ligand interpenetrate the receptor — real interfaces relieve this with side-chain motion — and $\rho^4 = 150$ per overlapping cell then falls only on the correct pose. A wrong pose that grazes the surface pays nothing. In the worst case (3wlb) the reachable native scores −4770.

For the remaining three the clash term is not the problem; the favourable pair count and the IFACE potential both prefer the wrong pose, pointing at D and at the pair table.

$\rho = 3.5$ is the value ZDOCK uses *with* an evenly distributed angle set and a downstream refinement stage. Without refinement the penalty is strong enough to eliminate the correct answer, which makes ρ — with its fourth-power sensitivity — the first parameter to calibrate, followed by α, D, and the pair table.

References

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